

Reflective Journal: Lab 08 Predicting Video Game Sales

Success with Machine Learning

The main thing that stuck with me while working through this lab was the importance of model iteration and evaluation. In particular, after reading Cell 13 and Cell 16 and starting to build the workflow, I began to think more about what I was doing and why. While going through earlier labs, I started to appreciate that executing cells in order is useful not just for reproducibility but for conceptual understanding. I kept this mindset for the remainder of the lab.

After visualizing the distribution of global and regional sales, the main takeaway for me was that games follow a long-tail distribution. There are going to be many more games that sell under a million copies than games that go on to sell millions or even tens of millions of copies. While creating the engineered feature in an earlier cell, I thought it would help but didn't realize it would end up being by far the most important feature. However, when creating our predictions () and probabilities () in Cell 19, it was evident that there was a strong relationship that was captured by this one feature.

The biggest surprise for me came when seeing the evaluation metrics in Cell 20. An accuracy of 0.9997, along with seeing the confusion matrix containing only 1 false classification was unbelievable to me. I honestly did not think we would be able to create a model that would classify every hit and only mislabel 1 game as a hit. Even the classification report enforced these feelings of us having a perfect model. But as I sat there looking at these results, I did not feel ecstatic. I felt confused — how is this possible? This caused me to think about things more, a skill that I have been trying to do more of. Is the model really that good? Or is this dataset made in a way that made the classification too easy? These are the types of thoughts I had while looking at these results.

Seeing the confusion matrix heatmap depicted in Cell 21 drove the story home for me. Seeing two big blocks along the diagonal and very little else was very interesting to look at. This cell along with the previous helped me visualize what was happening. Seeing a plot really can make things clearer, and this was something I came to appreciate in this cell.

While working through this lab, I ran into some small issues that stuck out to me. The first issue that I can remember is forgetting to place a in one of my cells. This caused the cell to execute and not show any output, because the notebook was kept in its state from previous executions. The second thing I noticed was that threw an error. However, this was because I ran the evaluation metrics cell before the cell that imported. These were both small things but things that stood out to me. Just need to slow down and make sure we take each step in the correct order.

I also initially evaluated the model in Cell 19, when we were supposed to leave that evaluation for Cell 20. Once I realized my mistake, I went back and recreated the cells so that predictions would be made in Cell 19 and the evaluations would be made in Cell 20. This helped me understand the flow of the lab better, as we always create our predictions first and then follow up with our evaluation.

The biggest takeaway that I can apply to business is that publishers can predict whether a game will be a hit or not by knowing the regional sales early on in a game's release. If a publisher knows the early sale numbers of a game, they can use the Cell 19 to predict the probability that a game will be a hit. If there are some games that are on the borderline of being classified as a hit, perhaps more efforts can be spent marketing the game or getting it more platform exclusive. Although our model was very simplistic, there are valuable lessons that can be applied to real situations.

The last thing I enjoyed about this lab was how it tied back to concepts that we learned in class. Decision Trees are said to be interpretable models, and this lab showed me why. Without even having to visualize the tree, we can infer from the evaluation metrics and how our feature behaved that there was a certain point where games were split into being a hit or not. We also learned about the trade-offs of decision trees, specifically overfitting. An accuracy of nearly 1 is likely overfitting to our data and may not classify new games (that this model has not seen) correctly.